FACULTY OF MECHANICAL ENGINEERING

CAN THO UNIVERSITY OF TECHNOLOGY

PHÁT TRIỂN ỨNG DỤNG IOT

(IoT application development)

Instructor : Dinh-Tu Nguyen (Ph.D.) – TS. Nguyễn Đình Tứ

Office: Faculty of Mechanical Engineering

Email: ndtu@ctuet.edu.vn

CHƯƠNG 4: CÁC NỀN TẢNG PHẦN CỨNG VÀ PHẦN MỀM IOT THÔNG DỤNG

4.1: Wireless sensor networks

4.2: Class of wireless sensor networks

4.3: The role of Open source in IoT

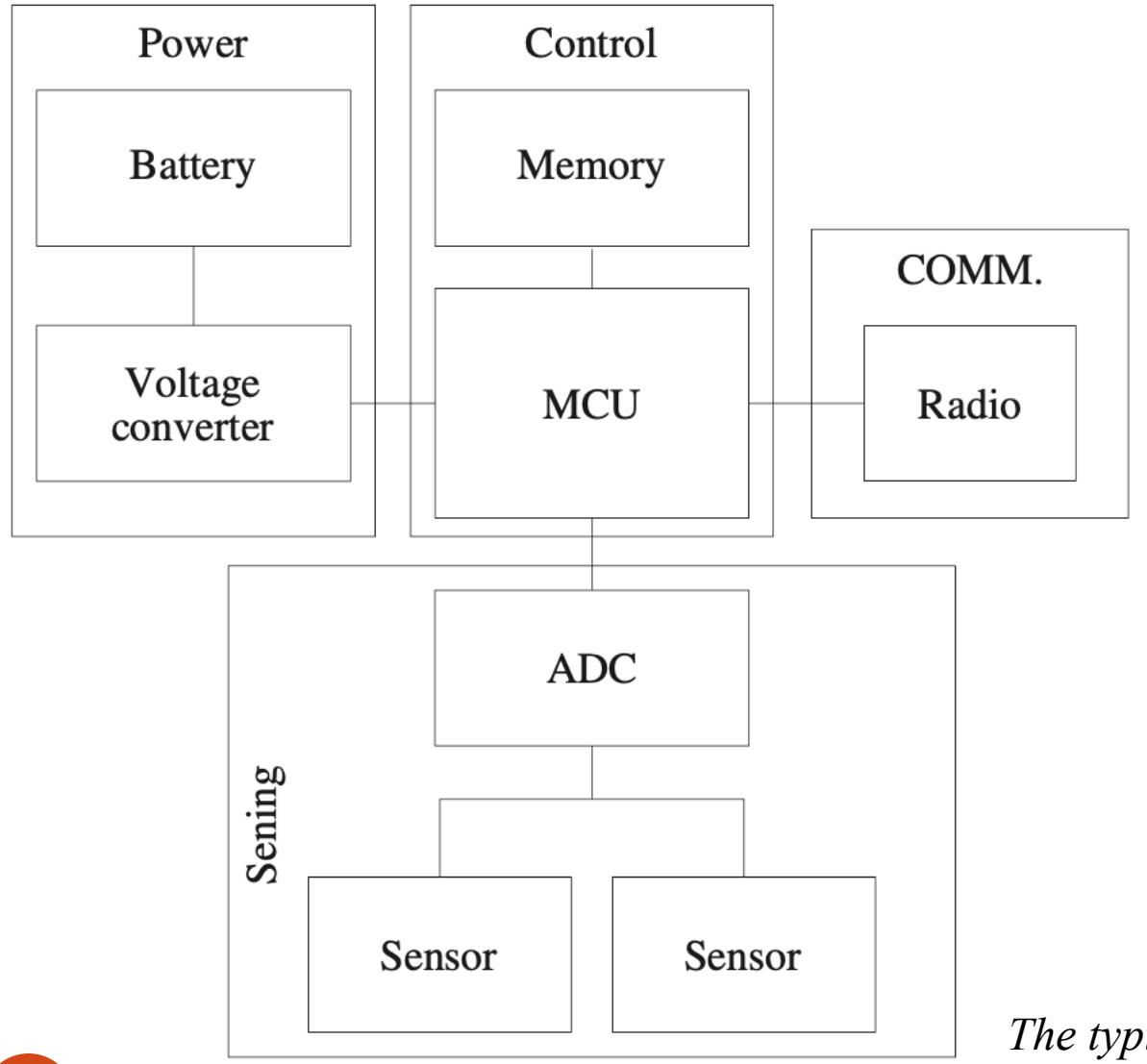
4.4: Các nền tảng phần cứng

4.5: Các nền tảng phần mềm

4.1. Wireless Sensor Networks

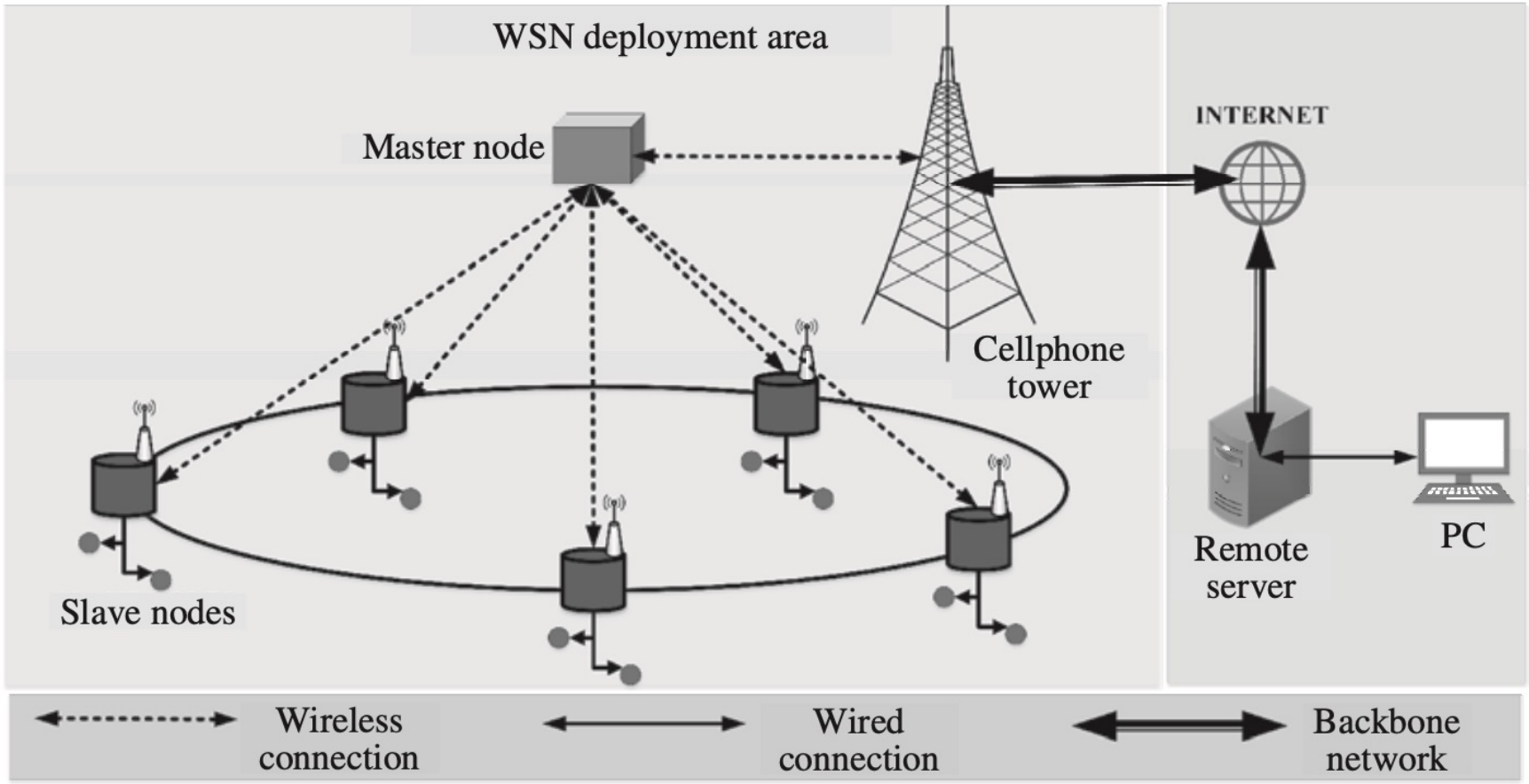
- **Wireless sensor networks** (**WSN**), as the name suggests, is a networking paradigm that makes use of spatially distributed sensors for gathering information concerning the immediate environment of the sensors and collecting the information centrally.
- Here, the sensors are not standalone devices but a combination of sensors, processors, and radio units—referred to as sensor nodes—sensing the environment and communicating the sensed data wirelessly to a remote location, which may or may not be connected to a backbone network.

4.1. Wireless Sensor Networks



The typical constituents of a WSN node

4.1. Wireless Sensor Networks



A typical WSN deployment

4.1. Wireless Sensor Networks

WSNs must have the following distinguishing features

- ✓ **Fault Tolerance**: The occurrence of faults in WSN nodes should not take down the whole WSN implementation, or hamper the transmission of data from non- faulty nodes to the central location.
- ✓ **Scalability**: WSN implementations must have the feature of scalability associated with their architectures and deployments. In the event of a future increase or decrease of sensor node units, the WSN must support the scaling of the infrastructure without changing the whole implementation.

4.1. Wireless Sensor Networks

- ✓ **Long lifetime**: The lifetime or the energy replenishment cycle of WSNs must be long enough to make large-scale applications feasible. WSNs have been used for monitoring remote, harsh, and hard to access environments; in these environments, it is not feasible to regularly replenish the energy source of the WSN nodes, which necessitates the need for long node lifetimes.
- ✓ **Security**: The security of WSNs, if not considered, can easily compromise the security of the whole system, right back to the central server. As WSNs are used for a wide range of applications, some of which are crucial, security is one aspect that must be properly addressed to prevent intrusion and maintain the integrity of the data.

4.1. Wireless Sensor Networks

- ✓ **Programmability**: The programmability of WSNs is important as it ensures the robustness of these systems. WSNs deployed in one application area can be reused for other applications just as easily with a change in sensors and the backend programs associated with it. Programmability also helps in providing a means of adjusting the parameters of the system in the event of a scale-up or scale-down operation.
- ✓ **Affordability**: As WSNs generally require multiple units, typically in the range of tens or hundreds of WSN nodes, the cost of the nodes and its affordability is vastly responsible for the acceptability of the system. Except for some specialized domains such as the military and the industry, where the sensing requirements are quite high and that too in harsh and challenging conditions, the majority of WSN applications are regular. To some extent, the cost of WSN deployments in these regular domains decide the acceptability of a WSN-based solution for that domain.

4.1. Wireless Sensor Networks

- ✓ **Heterogeneity**: The WSNs must support a wide number and various types of sensors and solutions, thus enabling heterogeneity. In the absence of heterogeneity, the WSN will tend to become very application-specific, which in turn would require major customizations even in the event of minor changes to the network or architecture.
- ✓ **Mobility**: WSNs must support the notion of mobility of nodes such that the nodes may be easily relocatable or mobile. Mobility would ensure the rapid deployability of WSN-based solutions in all environments types.

4.2. Class of Wireless Sensor Networks

- ***Military Applications***: WSNs are used for the detection of enemy soldiers, vehicles, intrusion, weapon systems, and armaments.
- ***Health Applications***: WSNs in healthcare are being used to monitor patients in hospitals, ambulances, and homes. Nowadays, a new class of healthcare devices—wearable appliances—enable a user to have a miniature health sensor on them without additional discomfort.
- ***Environmental Applications***: WSNs are used for environmental monitoring of pollution, tracking of wildlife, forests, and others.
- ***Home Applications***: WSNs in the home have given rise to home automation systems and smart home connectivity systems.

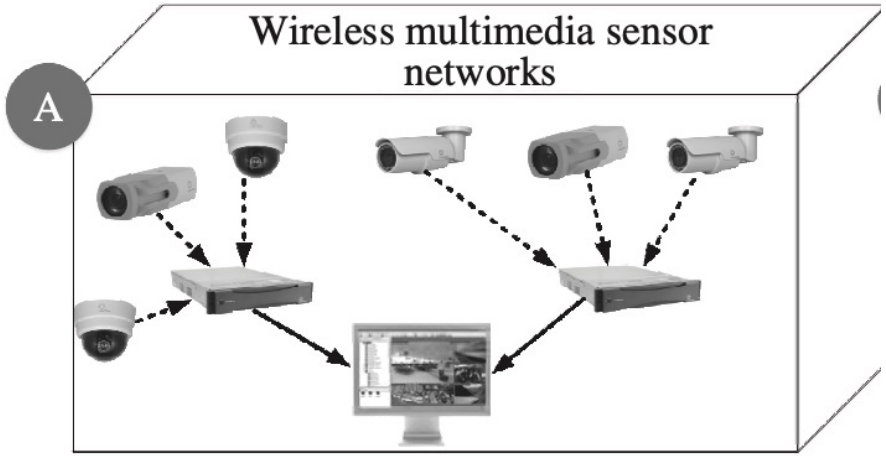
4.2. Class of Wireless Sensor Networks

- ***Commercial Applications***: WSNs are used for tracking of vehicles, packages in transport, logistics, and others.
- ***Industrial Monitoring***: WSNs in industries keep track of various industrial processes, monitor factory floors, ensure worker safety, and perform stock management.

4.2. Class of Wireless Sensor Networks

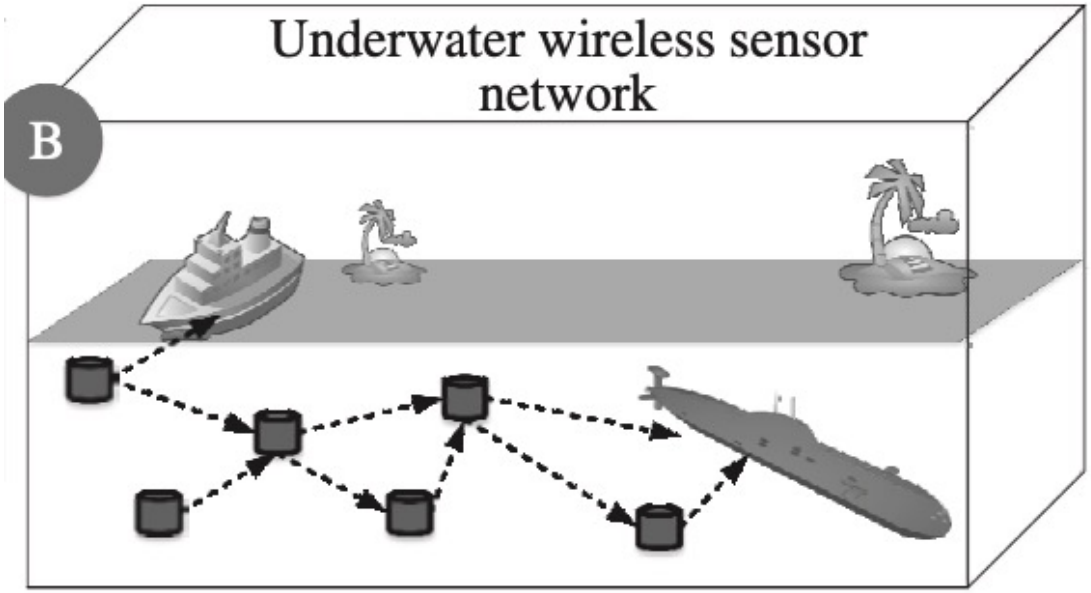
WSNs can be organized broadly into the following domains of implementation:

- **Wireless Multimedia Sensor Networks (WMSN)**: This class of WSNs boasts of the ability to retrieve videos, audios, images, or all three in addition to regular scalar sensor readings. The sensing range and coverage area of a camera- based WMSN are defined by the *field of view* (FOV) of the constituent cameras. Because of their superlative capabilities, WMSNs are popularly sought after in critical domains such as surveillance and road traffic monitoring. However, due to the use of multimedia sensors, the power and processing requirements of this class of WSNs are very high as compared to the other classes of WSNs. Additionally, WMSNs typically demand high network bandwidths for convenient operation.



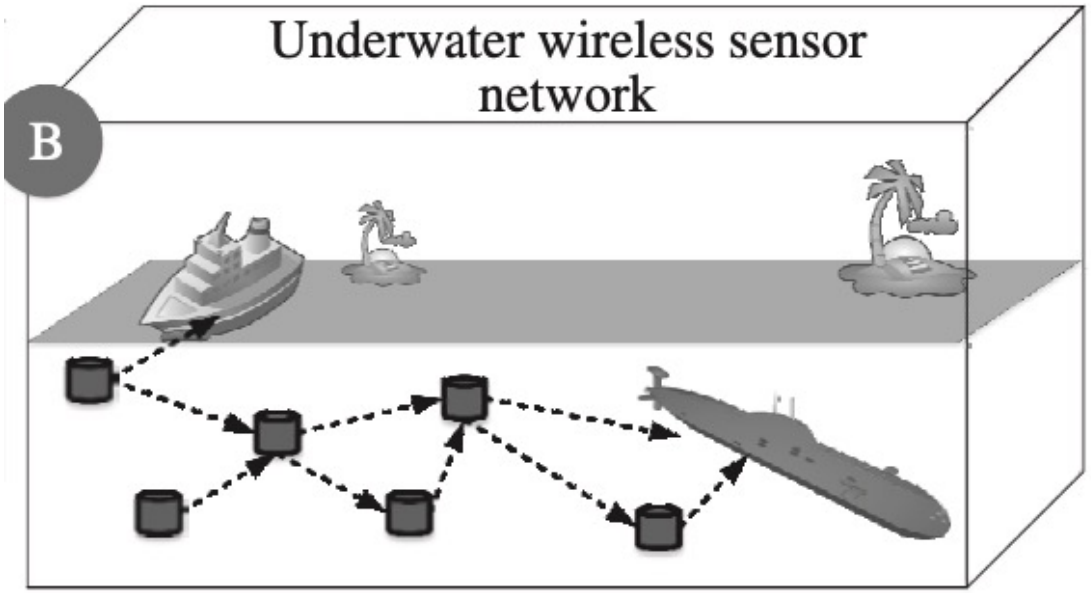
4.2. Class of Wireless Sensor Networks

- **Underwater Sensor Networks (UWSN)**: This class of WSN is designed specifically to work in underwater environments. Compared to terrestrial or aerial environments, wireless underwater communications are severely restricted in terms of data rate, range, and bandwidth due to the high underwater attenuation of electromagnetic signals. The reliable use of light- based underwater communication is also limited in terms of range and noise.



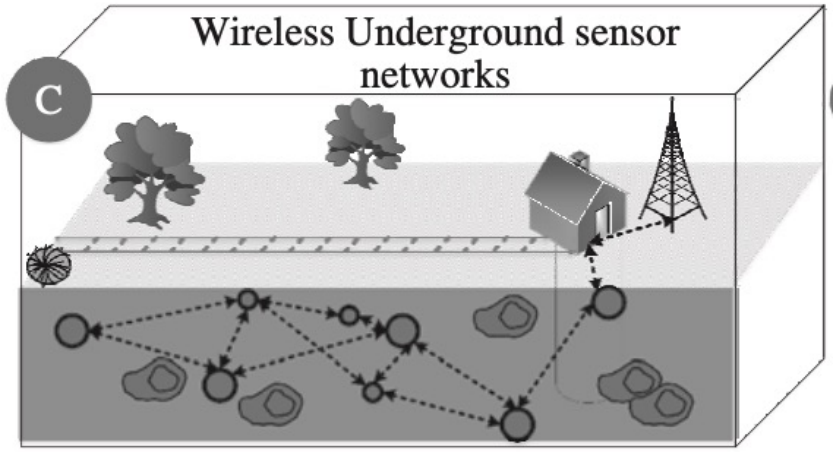
4.2. Class of Wireless Sensor Networks

The most accepted global means of underwater communication is by acoustic waves. However, the long propagation delays and uneven data rate makes it necessary to develop newer topologies and architectures, which can work under the conditions of severe limitations of the physical layer.



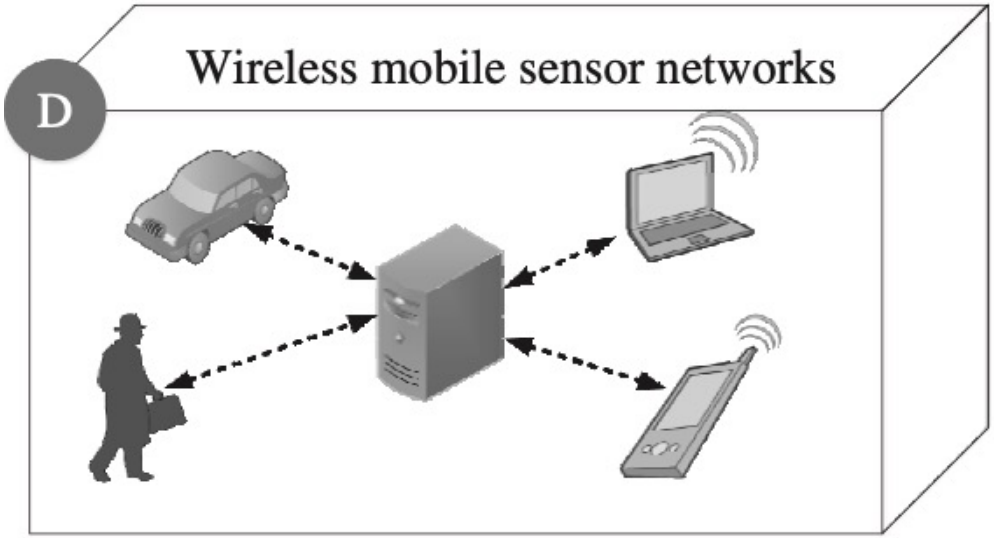
4.2. Class of Wireless Sensor Networks

- **Wireless Underground Sensor Networks (WUSN)**: This class of WSNs is designed to be deployed entirely underground. The underground environment poses challenges of attenuation due to the rocks and minerals in the soil. Another significant problem associated with this class is the need for digging up of the nodes to replenish their energy sources. Typical usage scenarios of this class of WSNs are underground mines and monitoring of underground plumbing systems. WUSNs need denser deployment architectures owing to the limited range of wireless communication in underground environments.



4.2. Class of Wireless Sensor Networks

❖ **Wireless Mobile Sensor Networks (MSN)**: This class of WSNs is characterized by its mobility and low power requirements. The sensor nodes are mobile, which requires them to rapidly connect to networks, disconnect from them, and then again connect to new networks until the nodes are mobile. Typical examples of MSNs include smartphone networks, wearables, vehicular networks, and others.



4.3. The Role of Open Source in IoT

The Open Source Movement

Open source in the computer industry is the publishing of source code or hardware design, with associated licensing that permits the reuse, modification, improvement, and potential commercialization under favorable terms. Example of favorable distribution terms includes the following criteria:

- ***Free Distribution***: Any party may sell or give away the open-source component as part of a larger system without being obligated to pay a royalty or other fee for such sale.
- ***Source Code/Design***: The source code or design must be distributed and made publicly available.
- ***Derived Works***: Derivation and modification of the original open-source component are allowed under the original licensing terms.

4.3. The Role of Open Source in IoT

The Open Source Movement

- While any system can potentially be released under an open source license by its owner, successful open source projects have associated communities of interest that are integral to their success.





4.4. Các nền tảng phần cứng

Thiết bị nhúng Arduino

- Board mạch Arduino đa dạng từ đơn giản tới những sản phẩm phục vụ thiết bị có thể mang theo tới các máy in 3D... Nhiều nhà nghiên cứu, sinh viên thực hiện các dự án với Arduino và cung cấp nhiều hướng dẫn, hỗ trợ online. Một số dòng thiết bị nổi tiếng như Arduino Yun, board Wi-Fi (IEEE 802.11 b/g/n) và Ethernet (IEEE 802.3 10/100Mb/s).



Arduino



Arduino IDE

4.4. Các nền tảng phần cứng

Thiết bị nhúng Arduino

- Board này sử dụng ATmega32, xung clock 16MHz, board này có 1 khe micro-SD, 1 port USB, 7 chan PWM pin, 12 analog input và 1 cổng micro USB.



Arduino



Arduino IDE

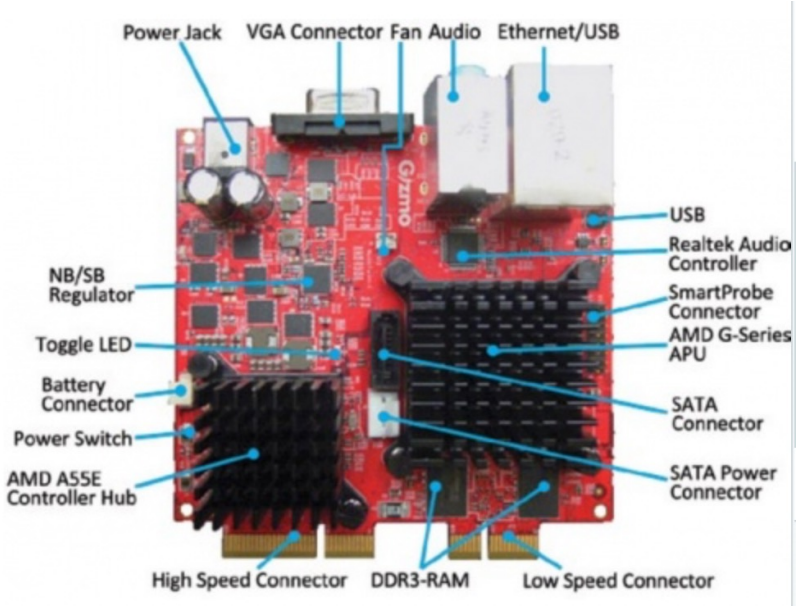
4.4. Các nền tảng phần cứng

IoT Devices- Hardware

- GizmoSphere is an open source development platform for the embedded design community; the site includes code downloads and hardware schematics along with free user guides, specification sheets, and other documentation.



GizmoSphere 2

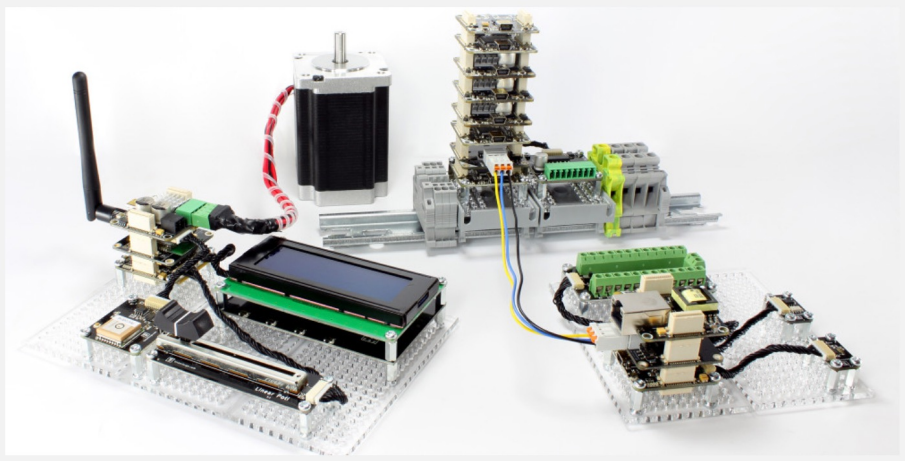


GizmoSphere

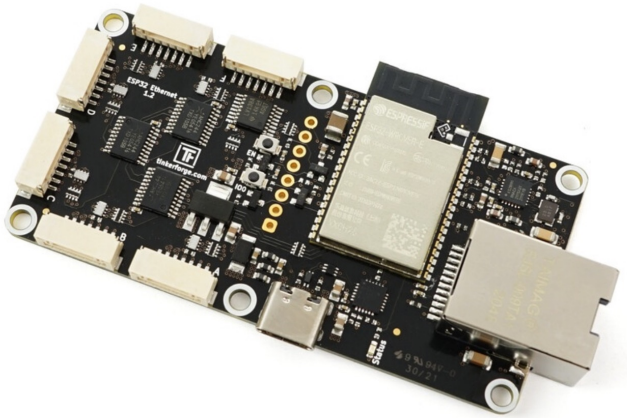
4.4. Các nền tảng phần cứng

IoT Devices- Hardware

- Tinkerforge is a system of open source stackable microcontroller building blocks. It allows the control of motors and reading out sensors with the following programming languages: C, C++, C#, Object Pascal, Java, PHP, Python, and Ruby over a USB or Wi-Fi connection on Windows, Linux, and Mac OS X. All of the hardware is licensed under CERN OHL (CERN Open Hardware License).



Tinkerforge

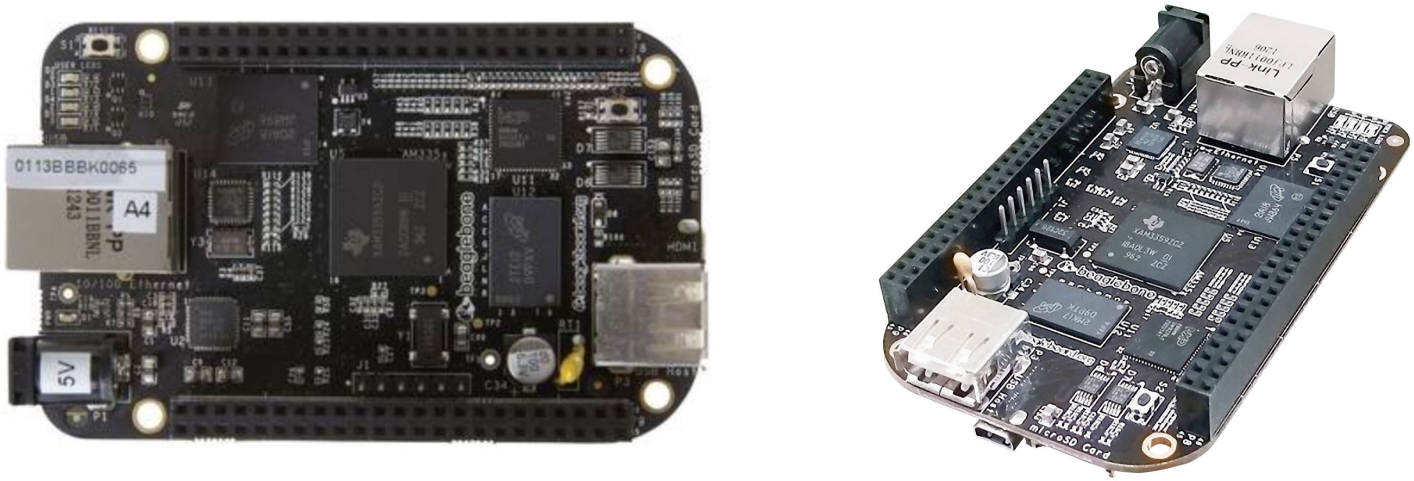


Tinkerforge stand-alone with ESP32 Bricks

4.4. Các nền tảng phần cứng

IoT Devices- Hardware

- BeagleBoard offers credit-card sized computers that can run Android and Linux. Because they have very low-power requirements, they are a good option for IoT devices. Both the hardware designs and the software they run are open source, and BeagleBoard hardware (often sold under the name BeagleBone) is available through a wide variety of distributors.



BeagleBone

4.4. Các nền tảng phần cứng

IoT Devices- Hardware

- As of 2020, the Raspberry Pi is the most used edge device for building IoT solutions among the DIY and maker communities. The affordable price and accessibility made it extremely popular. The computing and storage capabilities of the Raspberry Pi enables people to use it as an edge device within the IoT projects they are building.



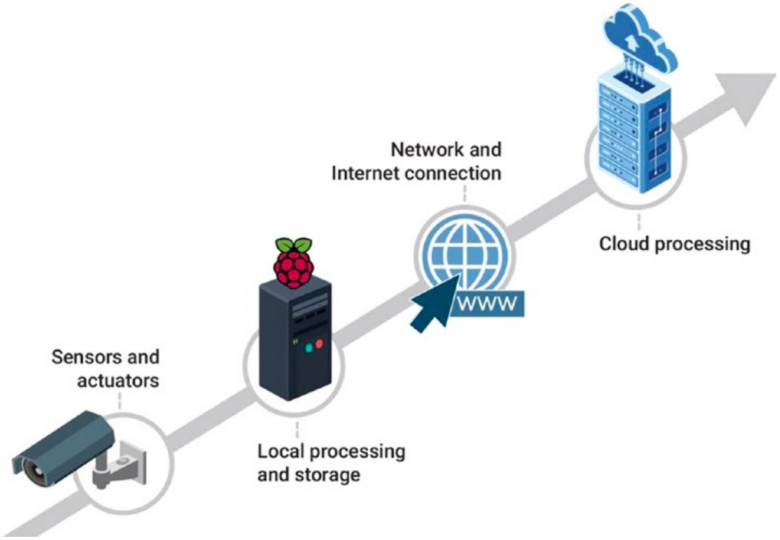
Raspberry Pi B+



Raspberry Pi Zero

4.4. Các nền tảng phần cứng

IoT Devices- Hardware



The Internet of Things stack

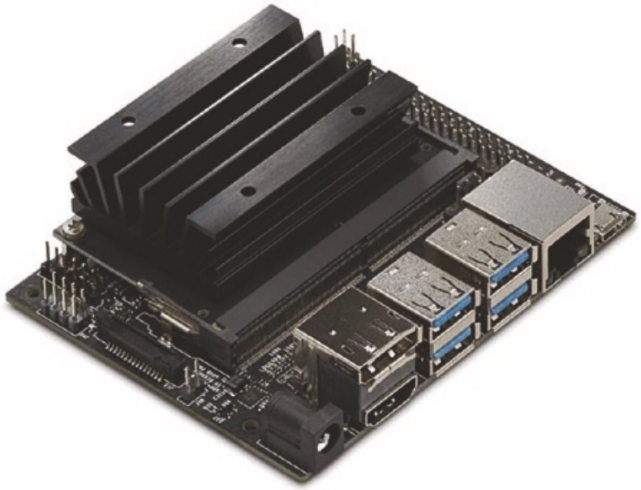
- Based on Figure, we notice that the sensors and actuators are placed at the bottom of the stack, as they are the ones communicating directly with the environment. Up the stack, we place the local processing and storage devices, also called edge devices.

- These are computers such as a Raspberry Pi, which are capable of exchanging information with the peripherals and have minimal computing capabilities so they can preprocess the data received from the sensors.

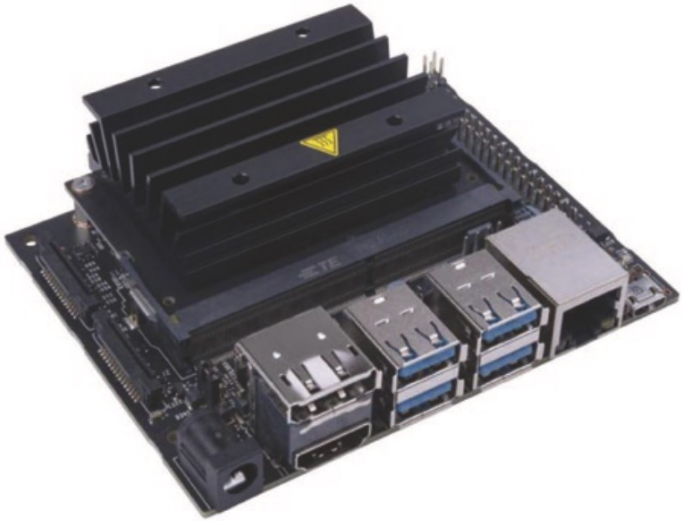
4.4. Các nền tảng phần cứng

IoT Devices- Hardware

- The Nvidia Jetson Nano was announced as a development system in mid-March of 2019. This product is intended for Internet of Things (IoT) makers. The board consists of a CPU with 1.43 GHz and a GPU with 128 cores of the



NVDIA Jetson Nano A02

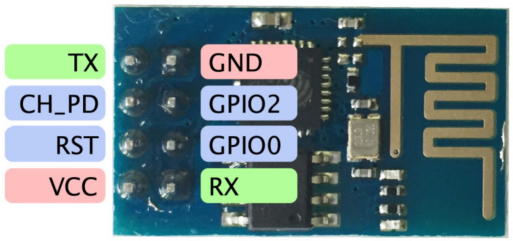


NVDIA Jetson Nano B01

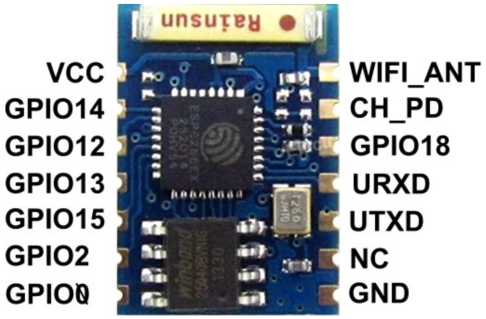
4.4. Các nền tảng phần cứng

IoT Devices- Hardware

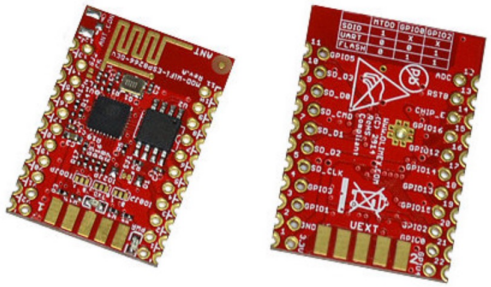
- ESP8266 is a wifi SoC (system on a chip) produced by Espressif Systems . It is an highly integrated chip designed to provide full internet connectivity in a small package. ESP8266 can be used as an external Wifi module, using the standard AT Command set Firmware by connecting it to any microcontroller using the serial UART, or directly serve as a Wifi-enabled microcontroller, by programming a new firmware using the provided SDK.



ESP8266-01



ESP8266-03

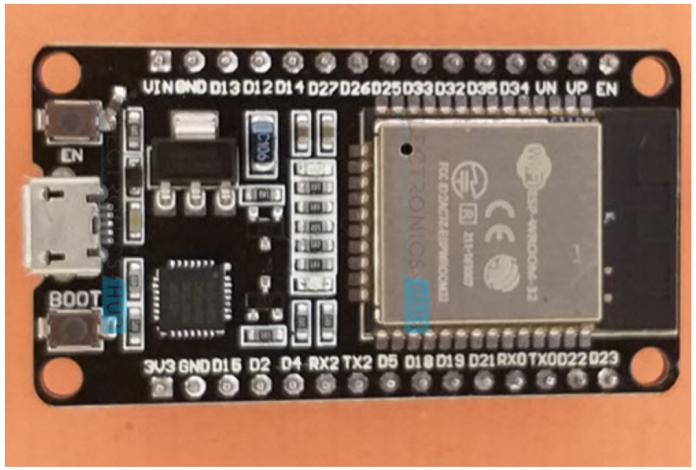


Olimex ESP8266

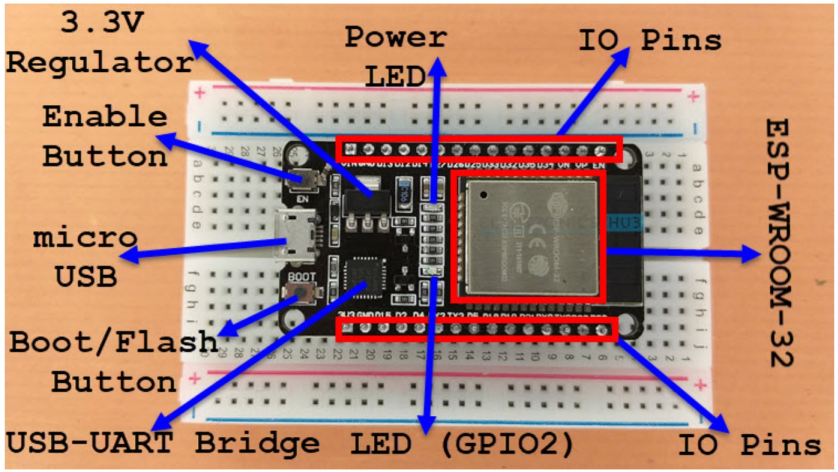
4.4. Các nền tảng phần cứng

IoT Devices- Hardware

- ESP32 is a low-cost System on Chip (SoC) Microcontroller from Espressif Systems, the developers of the famous ESP8266 SoC. It is a successor to ESP8266 SoC and comes in both single-core and dual-core variations of the Tensilica's 32-bit Xtensa LX6 Microprocessor with integrated Wi-Fi and Bluetooth.



ESP32 Devkit board

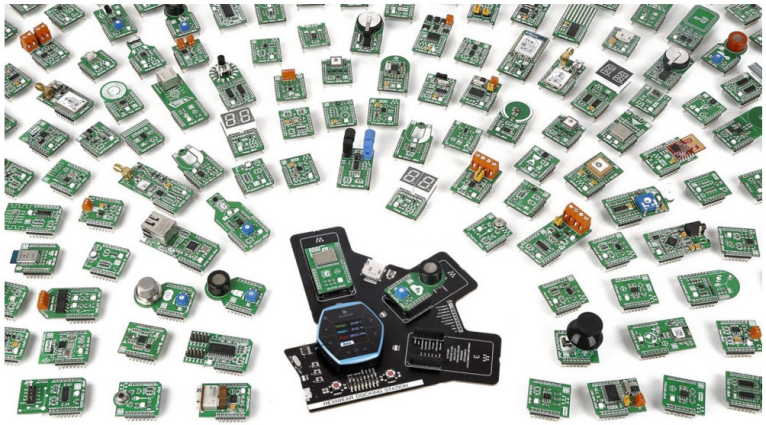


ESP32 Devkit board with its description

4.4. Các nền tảng phần cứng

IoT Devices- (wearables + IoT)

- The Zerynth IoT Platform is a full set of hardware-software tools designed by IoT experts to enable an easy transition from the prototyping phase to mass production in a flexible, scalable, and efficient way.





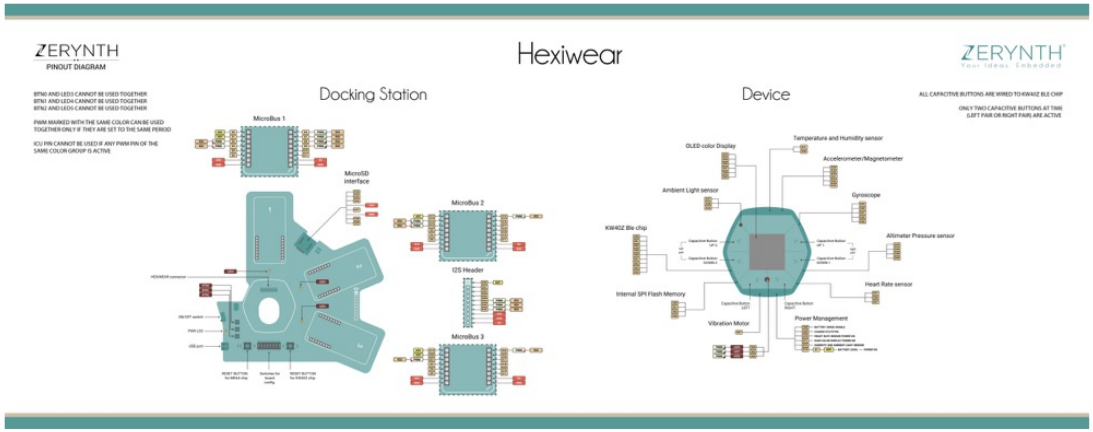
Zerynth IoT platform

4.4. Các nền tảng phần cứng

IoT Devices - (wearables + IoT)

- Zerynth OS is an operating system designed to simplify the development process. It features a Python/C Development environment and seamless integration with other hardware vendors.
- The Zerynth Cloud makes it possible to manage IoT devices, store data, visualize data, and gain insights from a single interface.



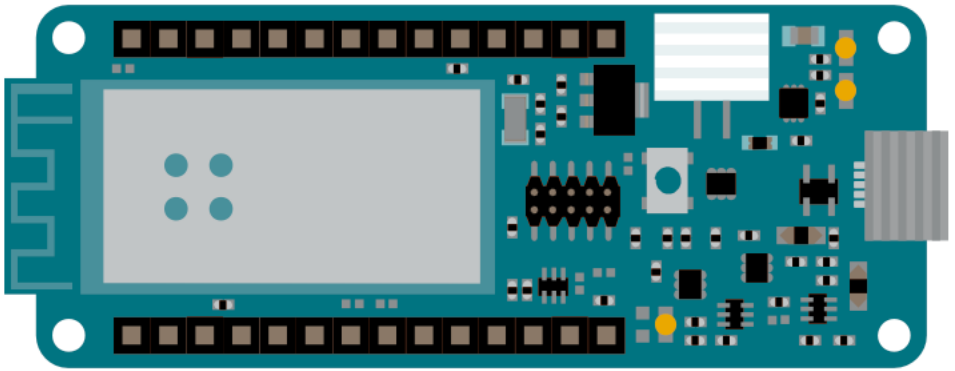


4.4. Các nền tảng phần cứng

IoT Devices - (wearables + IoT)

Alzheimer's is an extremely destructive disease that destroys the very essence of oneself by degrading memory and thinking abilities. Because 1 in 9 people aged 65 and above suffer from it, Abdullah Sadiq decided to build an assistive smartwatch that displays basic reminders, shows the time, monitors the wearer's location, and can even alert caretakers in the event of an emergency. It works by using an Infineon Sensor Hub in conjunction with an Arduino MKR1000 board to read sensor information and send it to either a smartphone or to an IoT platform.



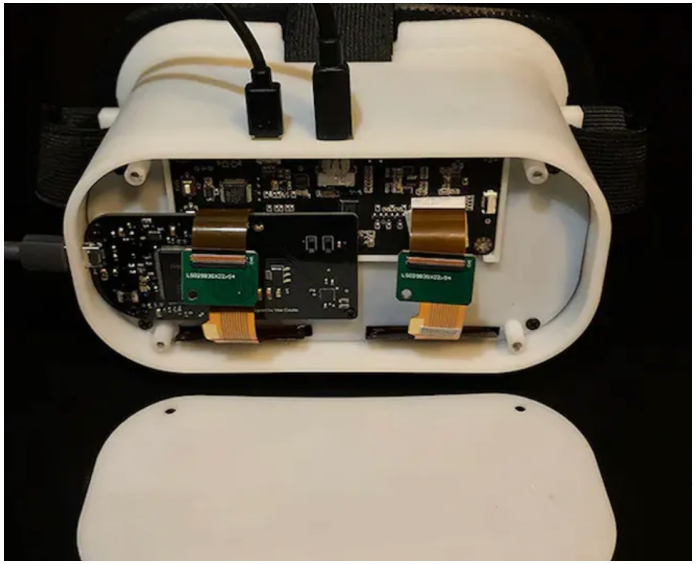


Arduino MKR1000 board

4.4. Các nền tảng phần cứng

IoT Devices - (Open source VR headset)

Probably the most advanced project in this list, the Relativity device is an open source VR headset that supports SteamVR and even body-tracking through the use of a neural network that estimates 3D positions. The headset was made possible thanks to its dual 2K displays that can output images at 120 fps. All of the computational horsepower comes from an Microchip SAM3X8E Cortex-M3 processor mounted to a custom PCB along with an MPU-6050 IMU for motion tracking.

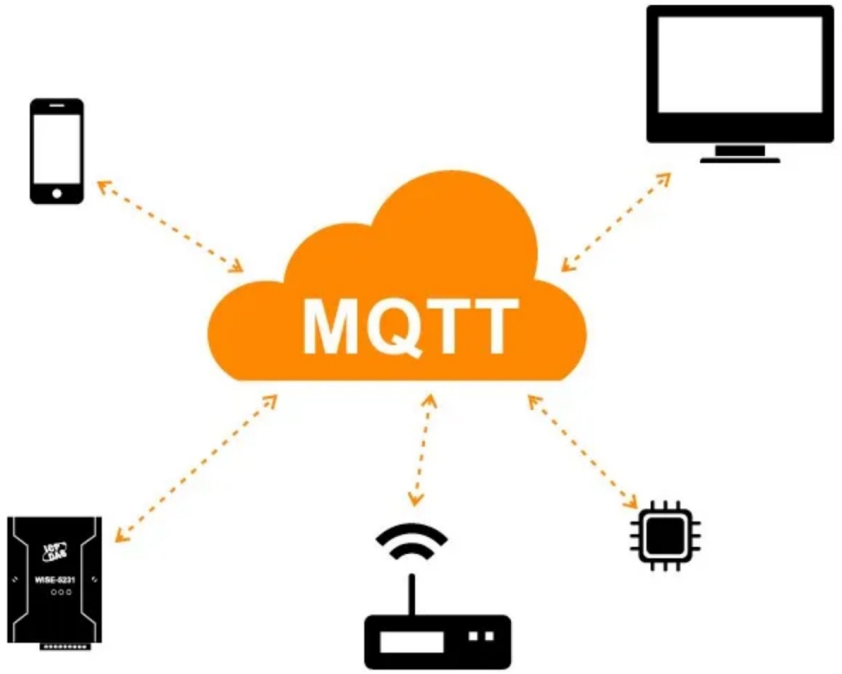


Open source VR headset

4.5. Các nền tảng phần mềm

Giao thức MQTT

- *MQTT* được thiết kế chủ yếu để kết nối các thiết bị bị hạn chết nguồn trên băng thông thấp. Mặc dù đã tồn tại trong hơn một thập kỷ nhưng chỉ khi có sự ra đời của M2M và IoT mới làm cho MQTT trở thành một giao thức phổ biến.



4.5. Các nền tảng phần mềm

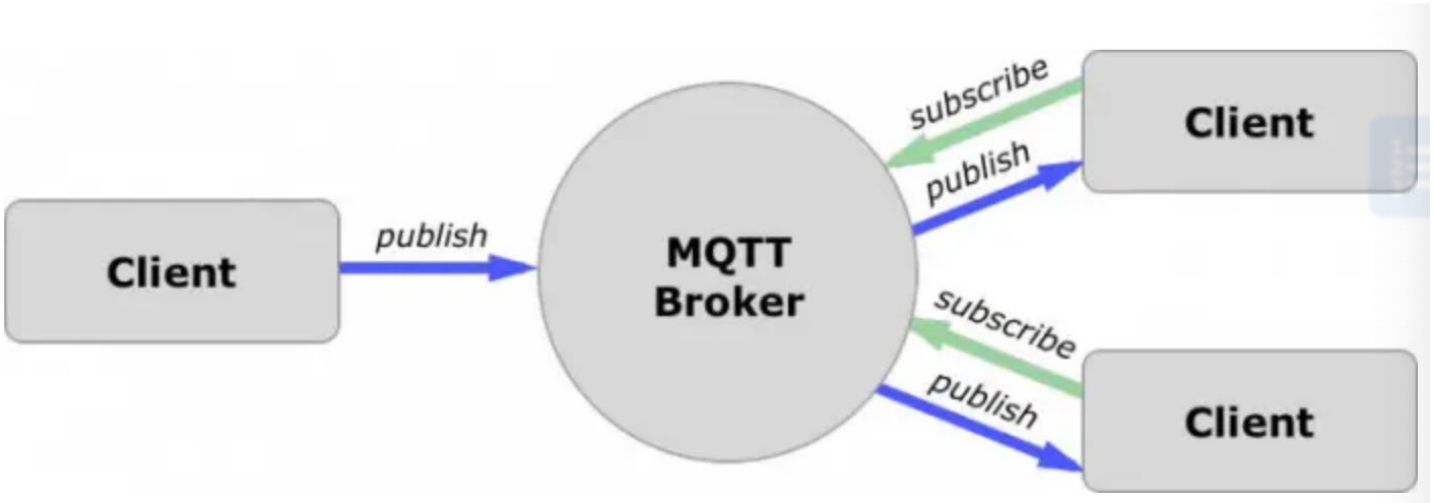
Giao thức MQTT

- *MQTT* (Message Queuing Telemetry Transport) là một giao thức truyền thông điệp (message) theo mô hình publish/subcribe (xuất bản – theo dõi), sử dụng băng thông thấp, độ tin cậy cao và có khả năng hoạt động trong điều kiện đường truyền không ổn định.
- MQTT gồm có các thành phần sau:
  - Client
  - Broker
  - Topic
  - Message
  - Publish
  - Subcribe
  - Connection

4.5. Các nền tảng phần mềm

Giao thức MQTT

- Giống như bất kỳ giao thức internet nào khác, MQTT dựa trên các client và một máy chủ. Tương tự như vậy, máy chủ (broker) là người chịu trách nhiệm xử lý các yêu cầu của client về việc nhận hoặc gửi dữ liệu giữa client và server với nhau.



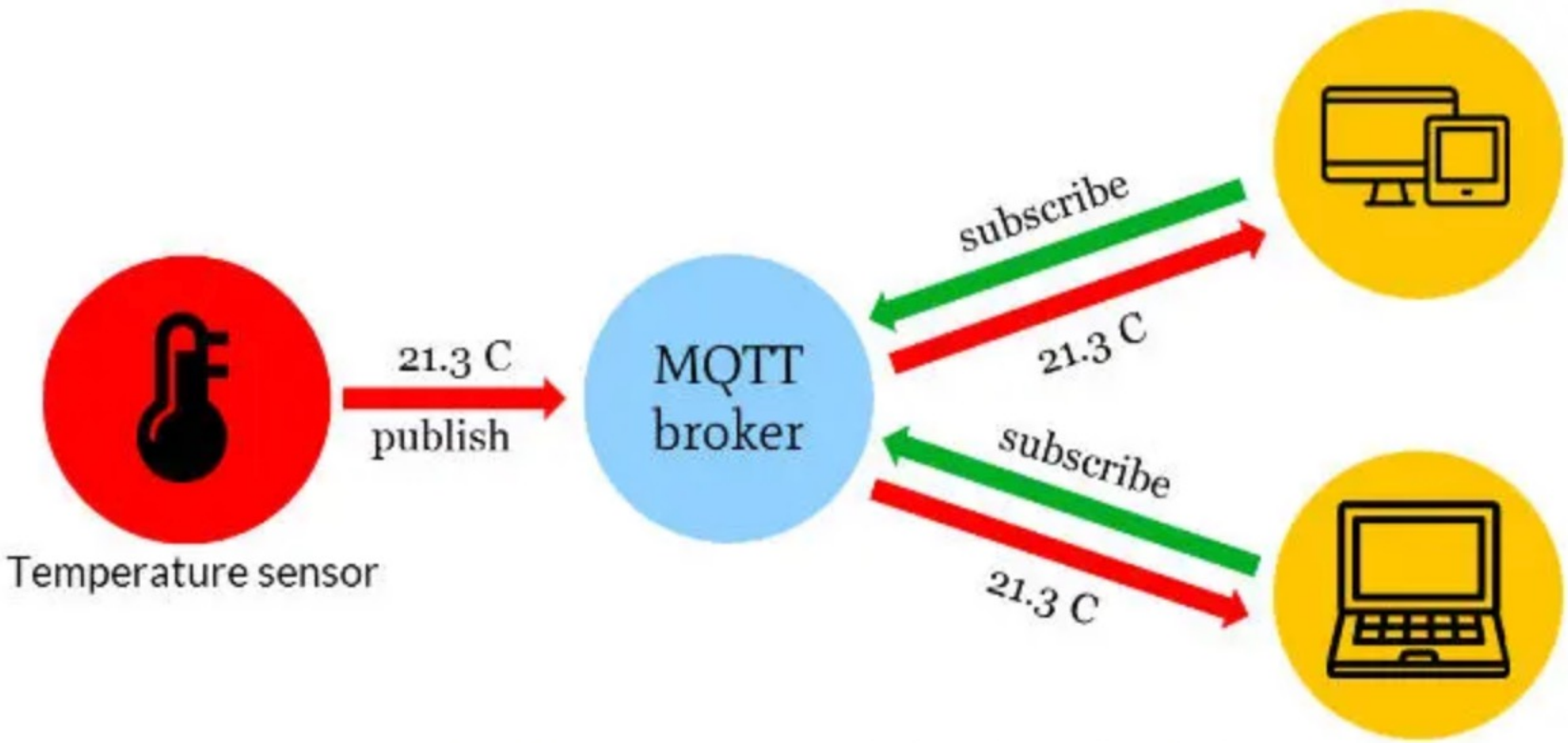
4.5. Các nền tảng phần mềm

Giao thức MQTT

- Giả sử có một thiết bị gắn cảm biến nhiệt độ. Nó sẽ gửi dữ liệu đọc được cho broker. Ở phía bên kia, ứng dụng của điện thoại hoặc PC nhận được giá trị của nhiệt độ này.
- Thiết bị xác định chủ đề (topic) mà nó muốn xuất bản (publish), ví dụ "temp". Sau đó, nó xuất bản thông báo "giá trị nhiệt độ".
- Ứng dụng điện thoại/máy tính PC đăng ký (subcribe) chủ đề "temp". Sau đó, nó nhận được thông báo rằng thiết bị đã xuất bản, đó là giá trị nhiệt độ.

4.5. Các nền tảng phần mềm

Giao thức MQTT



4.5. Các nền tảng phần mềm

Giao thức MQTT

- Có nhiều broker triển khai giao thức MQTT. Một trong những broker phổ biến nhất và thường được sử dụng là ***Mosquitto broker***.



4.5. Các nền tảng phần mềm

Giao thức MQTT

- Mosquitto là một broker thông điệp nguồn mở thực hiện giao thức MQTT. Nó gọn nhẹ và phù hợp để sử dụng trên tất cả các thiết bị từ một bo mạch đơn giản như Arduino, ESP8266 và ESP32 đến máy tính và máy chủ đầy đủ.
- Nhưng thay vì sử dụng Mosquitto trên máy tính cục bộ, chúng ta sẽ cần phải sử dụng một máy chủ dựa trên đám mây để triển khai broker Mosquitto. Điều đó là cần thiết để làm cho các dự án IoT có thể kiểm soát được trên Internet.
- Có rất nhiều broker mosquitto dựa trên đám mây như: ThingMQ, ThingStudio, MQTT.io, Heroku, CloudMQTT.

4.5. Các nền tảng phần mềm

Blynk

- *Blynk* là một nền tảng với các ứng dụng iOS và Android để điều khiển Arduino, Raspberry Pi và các ứng dụng tương tự qua Internet.
- Blynk là một bảng điều khiển kỹ thuật số nhờ đó bạn có thể xây dựng giao diện đồ họa cho dự án của mình bằng cách kéo và thả các widget.



- Blynk không bị ràng buộc với một số board mạch hoặc shield cụ thể. Thay vào đó, nó hỗ trợ phần cứng mà bạn lựa chọn. Cho dù Arduino hoặc Raspberry Pi của bạn được liên kết với Internet qua Wi-Fi. Blynk sẽ giúp bạn online và sẵn sàng cho IoT.

4.5. Các nền tảng phần mềm

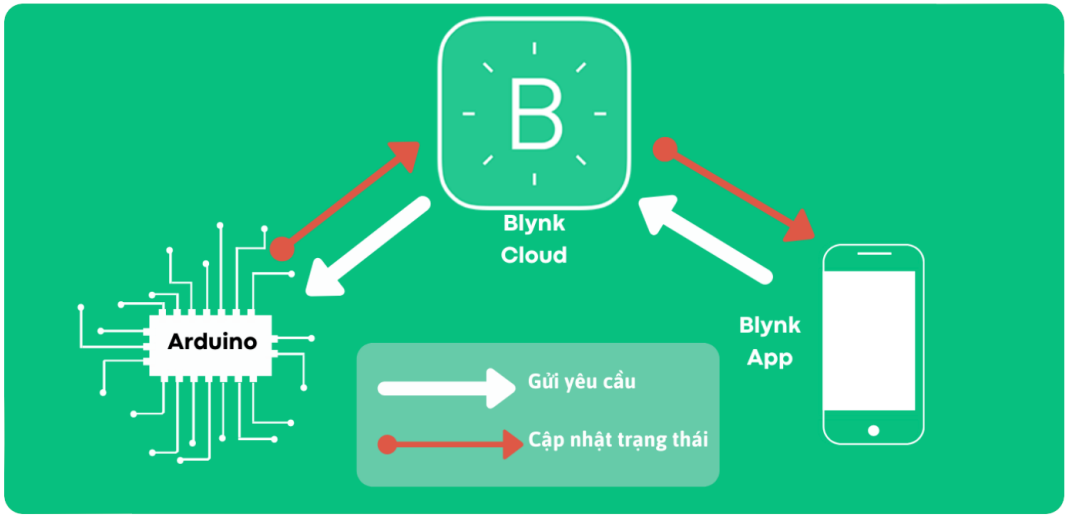
Blynk có thể làm được những gì?

- Triển khai một hệ thống nhà thông minh, điều khiển giám sát từ xa, v.v... Giới hạn duy nhất là khả năng cũng như sự sáng tạo của bạn.
- Các Widget như: nút ấn, hiện thị giá trị, thanh kéo giá trị, vẽ sơ đồ, v.v... Cho phép điều khiển GPIO của thiết bị bo mạch, từ đó bật tắt relay, điều khiển các thiết bị trong nhà.
- Điểm mạnh của Blynk là có thể kết nối ở bất kỳnơi nào có mạng.
- Có thể chia sẻ bảng điều khiển cho nhiều người dùng để sử dụng chung 1 hệ thống.

4.5. Các nền tảng phần mềm

Blynk có thể làm được những gì?

- Thiết bị arduino chịu trách nhiệm điều khiển các thiết bị trong nhà thông qua relay lắp trực tiếp vào các GPIO của arduino, hoặc bất kì kiểu giao tiếp nào khác nếu bạn đủ khả năng (RF, Uart). Thiết bị này yêu cầu phải kết nối mạng (ESP8266, ESP32, …)
- App Blynk từ điện thoại thông qua đám mây sẽ gửi tín hiệu đến board và ngược lại, các board arduino được đánh dấu bằng 1 token - chữ ký số, tạo ra trong quá trình setup ứng dụng.



Thanks for your listening!

